

Punya Mittal

Software Engineer | AI/ML | Backend Systems

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Professional Summary

Software Engineer and B.Tech. CS (AI) student with experience in building **backend systems, REST APIs, data pipelines, and machine learning systems**. Skilled in **Python, Java, Node.js, SQL, machine learning, and distributed systems**, with hands-on experience owning end-to-end engineering workflows.

Education

Vellore Institute of Technology (VIT), Chennai

B.Tech. Computer Science and Engineering (Artificial Intelligence)

2024 – 2028

CGPA: 8.55 / 10

Technical Skills

- **Languages:** Python, Java, JavaScript, TypeScript, SQL
- **Core CS:** Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design, Distributed Systems
- **Backend/Web:** Node.js, Express.js, FastAPI, React, Next.js, REST APIs, WebSockets, WebRTC, Microservices
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, NLP, RAG, Feature Engineering, AutoML
- **Data/Infra:** Pandas, NumPy, PostgreSQL, MySQL, MongoDB, Supabase, Redis, Docker, AWS, Linux, Git, CI/CD

Experience

Full Stack Engineering Intern

Apr 2026 – Present

JBN Technologies Pvt. Ltd.

- Manage the engineering of **CareerGUIDE**, an AI-driven career platform spanning frontend development, backend APIs, database architecture, assessment workflows, and AI-assisted career intelligence.
- Built an **O*NET ETL pipeline** processing **1,016 occupations, 57,543 alternate titles, and 220,098 ratings**; developed MBS-based assessment and scoring workflows for personalized career recommendations.

AI & Backend Engineering Intern

Oct 2025 – Jan 2026

Analytix4t Lab Pvt. Ltd.

- Designed and implemented **10+ REST APIs** in Node.js for financial analytics workflows spanning backend services, database operations, and application integrations.
- Engineered backend components for data processing, API integration, debugging, and performance optimization in production software.

AI/ML Engineering Intern

May 2025 – Jul 2025

ANNAM.AI, IIT Ropar

- Built end-to-end ML pipelines covering preprocessing, feature engineering, model training, and evaluation using Python, Pandas, and Scikit-learn.
- Trained and evaluated **XGBoost and LightGBM** models for agricultural intelligence and crop-yield prediction.

Selected Projects

Quorum — Semantic Merge Conflict Detection

Python, LLMs, Software Engineering

- Built an AI system to detect semantic conflicts between independently valid code changes; designed a multi-model adjudication pipeline and evaluated the system on a **300-pair leakage-free blind benchmark**, achieving **99.0% accuracy and 0.98 MCC**.

AutoML Pipeline — Natural Language to Deployed Model

Python, ML, Automation

- Built an automated pipeline that translates natural-language objectives into preprocessing, feature engineering, model selection, hyperparameter tuning, evaluation, and deployment across Random Forest, XGBoost, and LightGBM.

GINGR — Real-Time Communication Platform

Node.js, WebRTC, AWS

- Built a real-time messaging and video communication platform using WebRTC, WebSocket signaling, and STUN/TURN infrastructure for peer discovery, session management, and real-time communication.

Leadership and Achievements

Founder, Youth Season of Code (Y-SoC) — Built an open-source ecosystem with **250+ members, 100+ projects, and 55+ mentors** across 10+ colleges; Co-Founder, HackFinity.

Secretary, IE(I), VIT Chennai | Technical Lead, VoiceIT Club | 2x Hackathon Winner | NPTEL Top 1% — Conservation Economics, IIT Kanpur